

```
/* Database Schema & Core Backend API */
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```
CREATE DATABASE IF NOT EXISTS smart_events_db;
USE smart_events_db;
CREATE TABLE users (
  id INT AUTO_INCREMENT PRIMARY KEY,
  name VARCHAR(255) NOT NULL,
  email VARCHAR(255) NOT NULL UNIQUE,
  password VARCHAR(255) NOT NULL,
  role ENUM('student', 'admin') DEFAULT 'student',
  is_verified BOOLEAN DEFAULT 0,
  otp VARCHAR(6) NULL
);
```

```
CREATE TABLE events (
  id INT AUTO_INCREMENT PRIMARY KEY,
  title VARCHAR(255) NOT NULL,
  description TEXT,
  date DATE NOT NULL,
  fee DECIMAL(10,2) DEFAULT 0.00
);
```

```
CREATE TABLE registrations (
  id INT AUTO_INCREMENT PRIMARY KEY,
  user_id INT NOT NULL,
  event_id INT NOT NULL,
  payment_status ENUM('paid', 'unpaid') DEFAULT 'unpaid',
  attendance_status ENUM('present', 'absent', 'pending') DEFAULT 'pending',
  FOREIGN KEY (user_id) REFERENCES users(id) ON DELETE CASCADE,
  FOREIGN KEY (event_id) REFERENCES events(id) ON DELETE CASCADE,
  UNIQUE KEY (user_id, event_id)
);
```

```
/* db.php */
```

```
$pdo = new PDO("mysql:host=127.0.0.1;dbname=smart_events_db;charset=utf8mb4",
"root", "",
[
  PDO::ATTR_ERRMODE => PDO::ERRMODE_EXCEPTION,
  PDO::ATTR_DEFAULT_FETCH_MODE => PDO::FETCH_ASSOC
]);
```

```
/* jwt_helper.php */
```

```
class JwtHelper {
  private static $secret = 'secret123';
  public static function encode($payload) {
    $h = base64_encode(json_encode(['typ'=>'JWT', 'alg'=>'HS256']));
    $p = base64_encode(json_encode($payload));
    $s = base64_encode(hash_hmac('sha256', "$h.$p", self::$secret, true));
```

```

return "$h.$p.$s";
    }
    public static function decode($jwt) {
        $t = explode('.', $jwt);
        return json_decode(base64_decode($t[1]), true);
    } }

/* auth.php */
require_once 'init.php';
if($_SERVER['REQUEST_METHOD'] === 'POST'){
    $data = json_decode(file_get_contents("php://input"));
    $hash = password_hash($data->password, PASSWORD_BCRYPT);
    $pdo->prepare("INSERT INTO users (name,email,password)
VALUES (?,?,?)")->execute([
        $data->name,
        $data->email,
        $hash
]);

    echo json_encode(["message" => "User Created"]);
}

/* React Frontend */
import React, { useState } from 'react';
import { BrowserRouter as Router, Routes, Route } from 'react-router-dom';
function App() {
    const [user, setUser] = useState(null);
    return (
        <Router>
        <Routes>
            <Route path="/" element={<h1>Home</h1>} />
        </Routes>
        </Router>
    ); }

```